Soru 1-3

Uyg 1

$$\nabla = \frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z$$ : Nabla operatörü

türev alıcı

Gradyan

$f(x,y,z)$ fonksiyonu olsun skaler (örn: $ax+by+cz$)

$$\nabla f(x,y,z) = \frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial y}\vec{e}_z$$ : Gradyant vektöreldir

$\nabla f(x,y,z) = \text{grad} f$

skaler → [grad] → vektör

Diverjans

$$\vec{A} = A_x\vec{e}_x + A_y\vec{e}_y + A_z\vec{e}_z$$

$$\nabla \cdot \vec{A} = \text{div}\vec{A} = \left(\frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z\right)\left(A_x\vec{e}_x + A_y\vec{e}_y + A_z\vec{e}_z\right)$$

$$\text{div}\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

vektör → [div] → skaler

$\vec{e}_x \cdot \vec{e}_x = 1$ $\quad \vec{e}_x \times \vec{e}_x = 0$

$\vec{e}_x \cdot \vec{e}_y = 0$ $\quad \vec{e}_x \times \vec{e}_y = \vec{e}_z$

$\vec{e}_y \times \vec{e}_x = -\vec{e}_z$

$\vec{e}_y \times \vec{e}_z = \vec{e}_x$

$x \to y \to z$

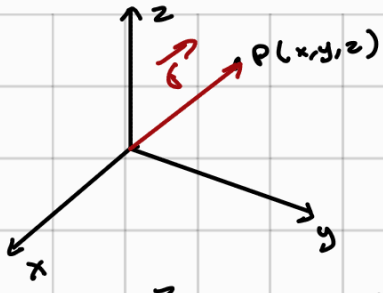
Rotasyonel

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

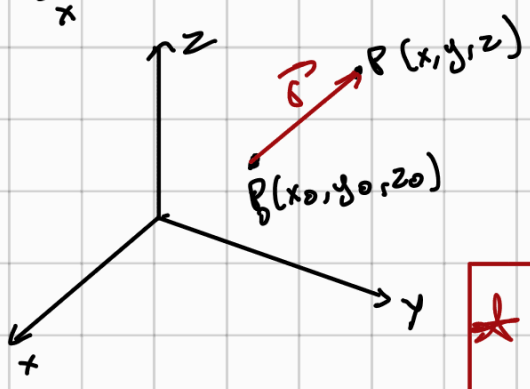
rot A

$$= \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right]\vec{e}_x + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right]\vec{e}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right]\vec{e}_z$$

vektör → [rot] → vektör




konum vektörü: $\vec{r} = x\vec{e}_x + y\vec{e}_y + z\vec{e}_z$




$$\vec{r} = (x-x_0)\vec{e}_x + (y-y_0)\vec{e}_y + (z-z_0)\vec{e}_z$$

$$|\vec{r}| = r = \sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}$$

$$\text{grad}\left(\frac{1}{r}\right) = ? = -\frac{\vec{r}}{r^3}$$

$$\text{grad}\left(\frac{1}{\sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2}}\right) = \frac{\partial}{\partial x}\frac{1}{r}\vec{e}_x + \frac{\partial}{\partial y}\frac{1}{r}\vec{e}_y + \frac{\partial}{\partial z}\frac{1}{r}\vec{e}_z$$

$$\frac{\partial}{\partial x}\frac{1}{r} = \frac{\partial}{\partial x}\left(\frac{1}{\sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2}}\right) = \left(\frac{\partial f(u)}{\partial u}\right)\frac{\partial u}{\partial x}$$

$u = (x-x_0)^2+(y-y_0)^2+(z-z_0)^2$

$f(u) = \frac{1}{\sqrt{u}}$

$$\frac{\partial f(u)}{\partial u} = \frac{\partial}{\partial u}\left(\frac{1}{\sqrt{u}}\right) = -\frac{1}{2}u^{-3/2}$$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2\right] = 2(x-x_0)$$

$$= -\frac{1}{2}u^{-3/2}\,2(x-x_0) = -\frac{x-x_0}{\left[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2\right]^{3/2}}$$

$$\frac{\partial}{\partial y}\frac{1}{r} = -\frac{y-y_0}{\left[(\text{---})\right]^{3/2}} \qquad \frac{\partial}{\partial z}\frac{1}{r} = -\frac{z-z_0}{\left[(\text{---})\right]^{3/2}}$$

$$\text{grad}\left(\frac{1}{r}\right) = -\underbrace{\frac{1}{\left[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2\right]^{3/2}}}_{-1/r^3}\underbrace{\left[(x-x_0)\vec{e}_x+(y-y_0)\vec{e}_y+(z-z_0)\vec{e}_z\right]}_{\vec{r}}$$

$$\text{grad}\left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$$

2) $\text{rot}\,\text{grad}\,f = 0$ Böyle olduğunu gösteriniz.

$f(x,y,z)$ skaler bir fonksiyon

$$\text{grad}\,f = \frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial z}\vec{e}_z$$

$$\text{rot grad}\,f = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} - \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix}$$

$$= \frac{\partial^2 f}{\partial y\partial z}\vec{e}_x + \frac{\partial^2 f}{\partial x\partial y}\vec{e}_z + \frac{\partial^2 f}{\partial z\partial x}\vec{e}_y - \underbrace{\frac{\partial^2 f}{\partial y\partial x}}_{=0}\vec{e}_z - \frac{\partial^2 f}{\partial z\partial y}\vec{e}_x - \frac{\partial^2 f}{\partial x\partial z}\vec{e}_y$$

3) $\vec{A} = A_x\vec{e}_x + A_y\vec{e}_y + A_z\vec{e}_z$

$\text{div}\,\text{rot}\,\vec{A} = 0$ olduğunu gösteriniz.

$$\text{rot}\,\vec{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right]\vec{e}_x + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right]\vec{e}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right]\vec{e}_z$$

$$\text{div}\,\text{rot}\,\vec{A} = \frac{\partial}{\partial x}\left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right] + \frac{\partial}{\partial y}\left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right] + \frac{\partial}{\partial z}\left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right]$$

$$= \frac{\partial^2 A_z}{\partial x\partial y} - \frac{\partial^2 A_y}{\partial x\partial z} + \frac{\partial^2 A_x}{\partial y\partial z} - \frac{\partial^2 A_z}{\partial y\partial x} + \frac{\partial^2 A_y}{\partial z\partial x} - \frac{\partial^2 A_x}{\partial z\partial y}$$

$\text{div}\,\text{rot}\,\vec{A} = 0$

4) $f(x,y,z)$ skaler fonksiyon $\vec{A}(x,y,z)$ vektör

$\text{div}\,(f.\vec{A}) = f\,\text{div}\vec{A} + \vec{A}\,\text{grad}f$ olduğunu gösteriniz.

$$\text{div}\,(f\vec{A}) = \frac{\partial}{\partial x}(f.A_x) + \frac{\partial}{\partial y}(f.A_y) + \frac{\partial}{\partial z} f(A_z)$$

$$= \left[f\frac{\partial A_x}{\partial x} + A_x\frac{\partial f}{\partial x}\right] + \left[f\frac{\partial A_y}{\partial y} + A_y\frac{\partial f}{\partial y}\right] + \left[f\frac{\partial A_z}{\partial z} + A_z\frac{\partial f}{\partial z}\right]$$

$$= f\left[\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right] + A_x\frac{\partial f}{\partial x} + A_y\frac{\partial f}{\partial y} + A_z\frac{\partial f}{\partial z}$$

$\left[\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right] = \text{div}\vec{A}$

$\text{grad}f = \frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial z}\vec{e}_z$

$$= f\,\text{div}\vec{A} + \vec{A}\,\text{grad}f$$

5) $\text{rot}\,(f\vec{A}) = f\,\text{rot}\vec{A} - \vec{A} \times \text{grad}f$ (× : cross product)

$f\vec{A} = fA_x\vec{e}_x + fA_y\vec{e}_y + fA_z\vec{e}_z$

$$= \left[\frac{\partial}{\partial y}(fA_z) - \frac{\partial}{\partial z}(fA_y)\right]\vec{e}_x + \left[\frac{\partial}{\partial z}(fA_x) - \frac{\partial}{\partial x}(fA_z)\right]\vec{e}_y$$

$$+ \left[\frac{\partial}{\partial x}(fA_y) - \frac{\partial}{\partial y}(fA_x)\right]\vec{e}_z$$

$$\text{rot}(f\vec{A}) = \left[f\frac{\partial A_z}{\partial y} + A_z\frac{\partial f}{\partial y} - f\frac{\partial A_y}{\partial z} - A_y\frac{\partial f}{\partial z}\right]\vec{e}_x$$

$$+ \left[f\frac{\partial A_x}{\partial z} + A_x\frac{\partial f}{\partial z} - f\frac{\partial A_z}{\partial x} - A_z\frac{\partial f}{\partial x}\right]\vec{e}_y$$

$$+ \left[f\frac{\partial A_y}{\partial x} + A_y\frac{\partial f}{\partial x} - f\frac{\partial A_x}{\partial y} - A_x\frac{\partial f}{\partial y}\right]\vec{e}_z$$

$\text{rot}\,\vec{A}$

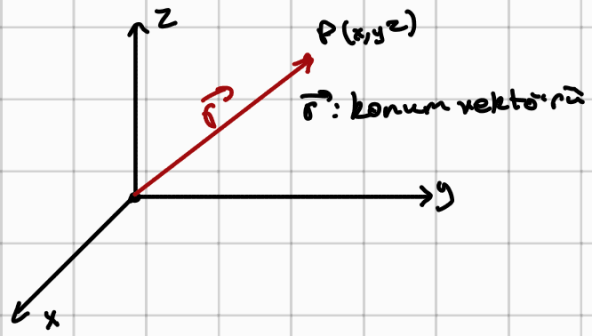
$$\text{rot}(f\vec{A}) = f\left[\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)\vec{e}_x+\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)\vec{e}_y+\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)\vec{e}_z\right]$$

$$+\left[\left(A_z\frac{\partial f}{\partial y}-A_y\frac{\partial f}{\partial z}\right)\vec{e}_x+\left(A_y\frac{\partial f}{\partial z}-A_z\frac{\partial f}{\partial x}\right)\vec{e}_y+\left(A_y\frac{\partial f}{\partial x}-A_x\frac{\partial f}{\partial y}\right)\vec{e}_z\right]$$

$-\vec{A}\times\text{grad}f$

$$= f\,\text{rot}\vec{A}-\vec{A}\times\text{grad}f$$

## Kartezyan koordinatlar




$$\vec{r} = x\vec{e}_x + y\vec{e}_y + z\vec{e}_z$$

$$|\vec{r}| = \sqrt{x^2+y^2+z^2}$$

$$h = \begin{bmatrix} h_x & 0 & 0 \\ 0 & h_y & 0 \\ 0 & 0 & h_z \end{bmatrix}$$ : metrik tensörü

$$\frac{\partial\vec{r}}{\partial x} = h_x\vec{e}_x = 1.\vec{e}_x \Rightarrow h_x = 1$$

$$\frac{\partial\vec{r}}{\partial y} = h_y\vec{e}_y \qquad h_y = 1$$

$$\frac{\partial\vec{r}}{\partial z} = h_z\vec{e}_z \qquad h_z = 1$$

C eğrisi olsun



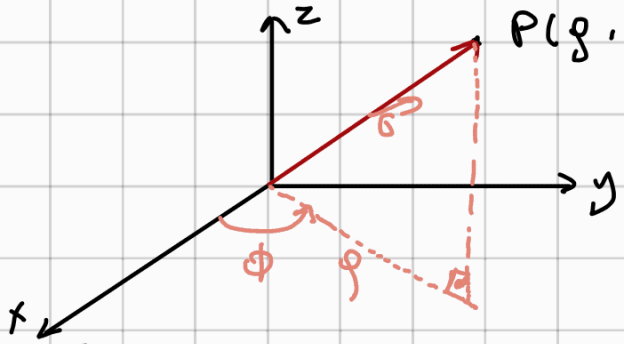

uzunluk → $d\vec{\ell} = h_x dx\vec{e}_x + h_y dy\vec{e}_y + h_z dz\vec{e}_z$

$$= dx\vec{e}_x + dy\vec{e}_y + dz\vec{e}_z$$

hacim → $dV = h_x h_y h_z\, dx\,dy\,dz$

$$= dx\,dy\,dz$$

## Silindirik koordinatlar




$$x = \rho\cos\phi$$
$$y = \rho\sin\phi$$
$$\vec{r} = \rho\cos\phi\,\vec{e}_x + \rho\sin\phi\,\vec{e}_y + z\,\vec{e}_z$$

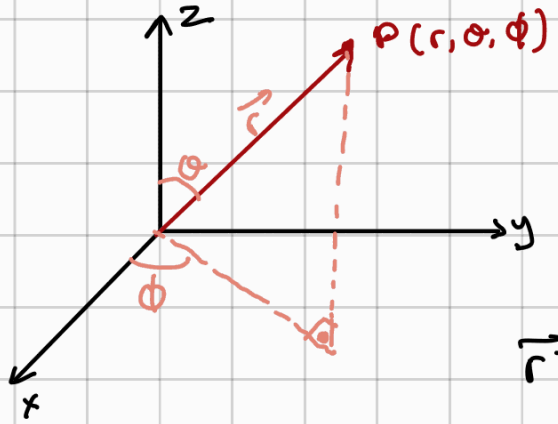
$$\begin{bmatrix} h_\rho & & \\ & h_\phi & \\ & & h_z \end{bmatrix}$$

$$\frac{\partial\vec{r}}{\partial\rho} = \cos\phi\,\vec{e}_x + \sin\phi\,\vec{e}_y = h_\rho\vec{e}_\rho$$
$$h_\rho = 1,\quad \vec{e}_\rho = \cos\phi\,\vec{e}_x + \sin\phi\,\vec{e}_y$$

$$\frac{\partial\vec{r}}{\partial\phi} = -\rho\sin\phi\,\vec{e}_x + \rho\cos\phi\,\vec{e}_y = h_\phi\vec{e}_\phi$$
$$h_\phi = \rho,\quad \vec{e}_\phi = -\sin\phi\,\vec{e}_x + \cos\phi\,\vec{e}_y$$

$$d\vec{\ell} = h_\rho\,d\rho\,\vec{e}_\rho + h_\phi\,d\phi\,\vec{e}_\phi + h_z\,dz\,\vec{e}_z$$
$$= d\rho\,\vec{e}_\rho + \rho\,d\phi\,\vec{e}_\phi + dz\,\vec{e}_z$$
$$dV = \rho\,d\rho\,d\phi\,dz$$

## küresel koordinatlar




$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$

$$\vec{r} = r\sin\theta\cos\phi\,\vec{e}_x + r\sin\theta\sin\phi\,\vec{e}_y + r\cos\theta\,\vec{e}_z$$

$$h_r = 1,\quad \vec{e}_r = \sin\theta\cos\phi\,\vec{e}_x + \sin\theta\sin\phi\,\vec{e}_y + \cos\theta\,\vec{e}_z$$
$$h_\theta = r,\quad \vec{e}_\theta = \cos\theta\cos\phi\,\vec{e}_x + \cos\theta\sin\phi\,\vec{e}_y - \sin\theta\,\vec{e}_z$$
$$h_\phi = r\sin\theta,\quad \vec{e}_\phi = -\sin\phi\,\vec{e}_x + \cos\theta\,\vec{e}_y$$

$$d\vec{\ell} = dr\,\vec{e}_r + r\,d\theta\,\vec{e}_\theta + r\sin\theta\,d\phi\,\vec{e}_\phi \qquad dV = r^2\sin\theta\,dr\,d\theta\,d\phi$$

$\varphi : 0 \to 2\pi$

$\sigma : 0 \to \pi$